Atlanta, Georgia
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Hunter Ellis

Electrical & Computer Engineer

ellishw.tech
github.com/hunterwellis
linkedin.com/in/ellishw

Electrical and Computer Engineer interested in the intersection of control systems, robotics, and autonomy.

Technical Experience

Avionics Research Engineer — Aug 2025 – Present
Georgia Tech Research Institute · Applied Systems Lab — *Atlanta, Georgia*

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Robotics Researcher / M.S. Computer Engineering — Aug 2023 – May 2025
Virginia Tech · COOL Autonomy Lab — *Blacksburg, Virginia*

- Built a 6-DOF robotic arm and accompanying ROS2–Gazebo & MuJoCo simulation environments for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.

Thrust Vector Control Intern — May 2024 – Aug 2024
Jacobs Space Exploration Group · Thrust Vector Control Lab — *Huntsville, Alabama*

- Developed thrust vector control testing hardware and software for NASA's Active Inertial Load (HIL) Simulator at the Marshall Space Flight Center.
- Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.

Control Research Intern — Jun 2023 – Aug 2023
G2ELab · Power Electronics Lab — *Grenoble, France*

- Researched microgrid inverter control systems – designed to be robust to islanding events and avoid future infrastructure problems on the French power grid.

Concept Design Intern — Jun 2022 – Aug 2022
Naval Research Enterprise Internship Program · NSWC Carderock — *West Bethesda, Maryland*

- Developed concept hospital sea-train designs and estimated fuel consumption and electrical power loads of the concept sea-trains.

Skills

Software: C/C++, Python, MATLAB, Simulink, GNU/Linux, Git, ROS2, Gazebo, Make, CMake, Labview, Qt, PyTorch, OpenCV, LaTeX, Verilog, FreeRTOS, Autodesk Inventor (Certified), SolidWorks, Rhino

Hardware: PCB Design and Assembly, Breadboarding, Computer Architecture, Oscilloscope, Multimeter, 3D-Printing, Hardware-in-Loop Simulation

Education

Master of Science in Computer Engineering — May 2025
Virginia Tech – Software and Machine Intelligence — *Blacksburg, Virginia*

Bachelor of Science in Electrical & Computer Engineering — May 2024
Virginia Tech – Control Systems and Machine Learning (double major) — *Blacksburg, Virginia*

Projects

6-DOF Robot Arm

- 3D printed robot arm, built using stepper motors and pulleys.
- Implemented ROS2 Jazzy control and Gazebo Harmonic simulation.

Papers

A Neuro-Symbolic Reinforcement Learning Architecture: Integrating Perception, Reasoning, and Control – Virginia Tech
Master's Thesis, Jun 2025